

Instructions

This mock mirrors the Infosys SP/DSE online assessment structure. Attempt it in one timed sitting; review using the highlighted answers.

Section 1 - Logical Reasoning | Section 2 - Quantitative Ability | Section 3 - Verbal Ability | Section 4 - Pseudocode/Technical | Section 5 - Coding (2 problems).

This is Mock Test 1 of 3; each mock uses a different question set.

Section 1 - Logical Reasoning

Q1. All cats are dogs. All dogs are animals. Conclusion: All cats are animals.

(A) Follows ✓

(B) Does not follow

(C) Cannot be determined

(D) Partially

Answer: (A) Follows

Authenticity: Frequently Repeated

Q2. No man is a woman. All women are humans. Conclusion: Some humans are not men.

(A) Follows ✓

(B) Does not follow

(C) Cannot be determined

(D) None

Answer: (A) Follows

Authenticity: Frequently Repeated

Q3. Next term: 3, 6, 9, 12, ?

(A) 14

(B) 15 ✓

(C) 16

(D) 18

Answer: (B) 15

Authenticity: Frequently Repeated

Q4. Letter series: A, C, E, G, ?

(A) H

(B) I ✓

(C) J

(D) K

Answer: (B) I

Authenticity: Frequently Repeated

Q5. If letters map to opposites (A<->Z, ...), CHAIR is coded as:

(A) XSZRI ✓

(B) XSAIR

(C) XSZIR

(D) XRZSI

Answer: (A) XSZRI

Authenticity: Community Reported

Q6. If A=1,B=2,..., value (sum of letters) of BOOK is:

(A) 41

(B) 43 ✓

(C) 45

(D) 40

Answer: (B) 43

Authenticity: Community Reported

Q7. P is the brother of Q, Q is the sister of R, R is the father of S. P is S's:

(A) Father

(B) Uncle ✓

(C) Brother

(D) Grandfather

Answer: (B) Uncle

Authenticity: Community Reported

Q8. My mother's brother's son is my:

(A) Uncle

(B) Nephew

(C) Cousin ✓

(D) Brother

Answer: (C) Cousin

Authenticity: Community Reported

Q9. Walk 5 km N, right 3 km, right 5 km. Distance from start:

(A) 5 km

(B) 3 km ✓

(C) 8 km

(D) 13 km

Answer: (B) 3 km

Authenticity: Community Reported

Q10. Walk 10 m S, left 10 m, left 10 m. Now facing:

(A) South

(B) East

(C) North ✓

(D) West

Answer: (C) North

Authenticity: Community Reported

Q11. Hand : Glove :: Foot : ?

(A) Shoe

(B) Sock ✓

(C) Toe

(D) Leg

Answer: (B) Sock

Authenticity: Community Reported

Q12. Book : Author :: Painting : ?

(A) Brush

(B) Painter ✓

(C) Canvas

(D) Colour

Answer: (B) Painter

Authenticity: Community Reported

Q13. Angle between clock hands at 3:00:

(A) 60 deg

(B) 90 deg ✓

(C) 120 deg

(D) 180 deg

Answer: (B) 90 deg

Authenticity: Community Reported

Q14. Odd days in 100 years:

(A) 4

(B) 5 ✓

(C) 6

(D) 0

Answer: (B) 5

Authenticity: Research-Backed Company Style

Section 2 - Quantitative Ability

Q1. Find the next term: 2, 6, 12, 20, 30, ?

(A) 36

(B) 40

(C) 42 ✓

(D) 44

Answer: (C) 42

Authenticity: Frequently Repeated

Q2. Find the next term (Fibonacci): 1, 1, 2, 3, 5, 8, ?

(A) 11

(B) 12

(C) 13 ✓

(D) 15

Answer: (C) 13

Authenticity: Frequently Repeated

Q3. Find 40% of 60% of 500.

(A) 100

(B) 120 ✓

(C) 150

(D) 200

Answer: (B) 120

Authenticity: Frequently Repeated

Q4. In an exam 35% failed in Maths, 45% in English and 20% in both. What percent passed both?

(A) 40% ✓

(B) 20%

(C) 60%

(D) 35%

Answer: (A) 40%

Authenticity: Frequently Repeated

Q5. Cost price Rs. 400, selling price Rs. 500. Profit percent:

(A) 20%

(B) 25% ✓

(C) 10%

(D) 15%

Answer: (B) 25%

Authenticity: Community Reported

Q6. If CP of 20 articles equals SP of 16 articles, profit percent:

(A) 20%

(B) 25% ✓

(C) 16%

(D) 30%

Answer: (B) 25%

Authenticity: Community Reported

Q7. A finishes in 10 days, B in 15 days. Together they finish in:

(A) 5

(B) 6 ✓

(C) 7.5

(D) 8

Answer: (B) 6

Authenticity: Community Reported

Q8. Pipe A fills in 6 h, B in 4 h. Together:

(A) 2 h

(B) 2.4 h ✓

(C) 3 h

(D) 2.5 h

Answer: (B) 2.4 h

Authenticity: Community Reported

Q9. Convert 90 km/h to m/s.

(A) 25 m/s ✓

(B) 30 m/s

(C) 20 m/s

(D) 15 m/s

Answer: (A) 25 m/s

Authenticity: Community Reported

Q10. Two trains (140 m, 160 m) in opposite directions at 60 and 40 km/h cross in:

(A) 10.8 s ✓

(B) 12 s

(C) 9 s

(D) 15 s

Answer: (A) 10.8 s

Authenticity: Community Reported

Q11. Divide Rs. 600 in the ratio 2 : 3. Smaller share:

(A) Rs. 200

(B) Rs. 240 ✓

(C) Rs. 260

(D) Rs. 300

Answer: (B) Rs. 240

Authenticity: Community Reported

Q12. Ratio 5:8; each increased by 4 becomes 2:3. The numbers:

(A) 20 and 32 ✓

(B) 15 and 24

(C) 25 and 40

(D) 10 and 16

Answer: (A) 20 and 32

Authenticity: Research-Backed Company Style

Q13. Simple interest on Rs. 5000 at 10% for 2 years:

(A) Rs. 1000 ✓

(B) Rs. 1100

(C) Rs. 500

(D) Rs. 1050

Answer: (A) Rs. 1000

Authenticity: Community Reported

Q14. Difference between CI and SI at 10% for 2 years is Rs. 50. The sum:

(A) Rs. 5000 ✓

(B) Rs. 4000

(C) Rs. 6000

(D) Rs. 4500

Answer: (A) Rs. 5000

Authenticity: Research-Backed Company Style

Section 3 - Verbal Ability

Q1. Synonym of ABUNDANT:

(A) Scarce

(B) Plentiful ✓

(C) Rare

(D) Empty

Answer: (B) Plentiful

Authenticity: Community Reported

Q2. Synonym of BENEVOLENT:

(A) Cruel

(B) Kind ✓

(C) Greedy

(D) Strict

Answer: (B) Kind

Authenticity: Community Reported

Q3. Antonym of TRANSPARENT:

(A) Clear

(B) Opaque ✓

(C) Bright

(D) Thin

Answer: (B) Opaque

Authenticity: Community Reported

Q4. Antonym of ANCIENT:

(A) Old

(B) Modern ✓

(C) Historic

(D) Aged

Answer: (B) Modern

Authenticity: Community Reported

Q5. Choose the correct sentence (subject-verb agreement):

(A) He do not like coffee.

(B) He does not likes coffee.

(C) He does not like coffee. ✓

(D) He not like coffee.

Answer: (C) He does not like coffee.

Authenticity: Community Reported

Q6. Choose the correct sentence (tense with 'yesterday'):

(A) I have visited him yesterday.

(B) I had visited him yesterday.

(C) I visited him yesterday. ✓

(D) I visit him yesterday.

Answer: (C) I visited him yesterday.

Authenticity: Community Reported

Q7. He is good ____ mathematics.

(A) in

(B) at ✓

(C) on

(D) with

Answer: (B) at

Authenticity: Community Reported

Q8. He succeeded ____ hard work.

(A) by ✓

(B) with

(C) from

(D) in

Answer: (A) by

Authenticity: Community Reported

Q9. Passage: 'Renewable energy sources such as solar and wind are increasingly important because they are clean and inexhaustible. Unlike fossil fuels they do not release harmful gases. However, their output depends on weather, making storage essential.' | Why are renewables important?

(A) Cheap to install

(B) Clean and inexhaustible ✓

(C) They work in all weather

(D) They replace storage

Answer: (B) Clean and inexhaustible

Authenticity: Research-Backed Company Style

Q10. (Same passage) Compared with fossil fuels, renewables:

- (A) release more gases
- (B) release gases equally

(C) do not release harmful gases ✓

- (D) cannot be used

Answer: (C) do not release harmful gases

Authenticity: Research-Backed Company Style

Q11. Order P,Q,R,S. P: It began to rain heavily. Q: We were playing in the park. R: So we ran for shelter. S: The sky suddenly turned dark.

(A) Q S P R ✓

- (B) P Q R S
- (C) S P Q R
- (D) Q P S R

Answer: (A) Q S P R

Authenticity: Research-Backed Company Style

Section 4 - Pseudocode / Technical

Q1. Time complexity of binary search on a sorted array of n elements:

- (A) $O(n)$

(B) $O(\log n)$ ✓

- (C) $O(n \log n)$
- (D) $O(1)$

Answer: (B) $O(\log n)$

Authenticity: Frequently Repeated

Q2. Worst-case time complexity of bubble sort:

- (A) $O(n)$
- (B) $O(n \log n)$

(C) $O(n^2)$ ✓

- (D) $O(\log n)$

Answer: (C) $O(n^2)$

Authenticity: Community Reported

Q3. A tree with n nodes has how many edges?

- (A) n

(B) n-1 ✓

- (C) n+1
- (D) 2n

Answer: (B) n-1

Authenticity: Community Reported

Q4. function f(n){ if(n<=1) return 1; return n*f(n-1); } f(4) returns:

(A) 12

(B) 24 ✓

(C) 16

(D) 6

Answer: (B) 24

Authenticity: Frequently Repeated

Q5. sum=0; for i in 1..n: for j in 1..n: sum++. Final sum:

(A) n

(B) 2n

(C) n^2 ✓

(D) $n \log n$

Answer: (C) n^2

Authenticity: Community Reported

Q6. a=[1,2,3,4]; s=0; for x in a: s+=x; print(s). Output:

(A) 6

(B) 10 ✓

(C) 24

(D) 4

Answer: (B) 10

Authenticity: Community Reported

Section 5 - Coding

Problem 1: Second Largest Element

Difficulty: **Easy** | Pattern: **Array Traversal** | Authenticity: **Community Reported**

Problem Statement

Find the second largest element in an array. Print -1 if it does not exist.

Input Format

Line 1: N. Line 2: N integers.

Output Format

The second largest element or -1.

Constraints

$1 \leq N \leq 10^5$

Sample Input

```
6
12 35 1 10 34 1
```

Sample Output

```
34
```

Python Solution

```
def second_largest(arr):
    largest = second = float('-inf')
    for x in arr:
        if x > largest:
            second, largest = largest, x
        elif largest > x > second:
            second = x
    return second if second != float('-inf') else -1

print(second_largest([12, 35, 1, 10, 34, 1]))
```

Step-by-Step Explanation

Track largest and second largest in one pass; a new maximum pushes the old maximum into second place.

Dry Run

35->largest;12->second;34 sits between 12 and 35->second=34. Result 34.

Time Complexity: **O(N)** | Space Complexity: **O(1)**

Problem 2: LCM of Two Numbers

Difficulty: *Easy* | Pattern: *Number Theory (GCD)* | Authenticity: *Community Reported*

Problem Statement

Compute the least common multiple of two positive integers.

Input Format

Two integers a and b.

Output Format

The LCM.

Constraints

$1 \leq a, b \leq 10^9$

Sample Input

4 6

Sample Output

12

Python Solution

```
import math
def lcm(a, b):
    return a * b // math.gcd(a, b)

print(lcm(4, 6))
```

Step-by-Step Explanation

$\text{LCM}(a,b) = a*b / \text{GCD}(a,b)$. $\text{GCD}(4,6)=2$, so $\text{LCM} = 24/2 = 12$.

Dry Run

gcd=2 -> lcm=12.

Time Complexity: **$O(\log(\min(a,b)))$** | Space Complexity: **$O(1)$**

Infosys

Full-Length Mock Test 1

Applied Role(s):

Digital Specialist Engineer (Trainee) | Specialist Programmer L1 / L2 / L3 (Trainee)

Online Assessment Preparation Material

Version 1.1 • Last Updated: July 2026

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.